

Predicting Employee Attrition Through Workforce Data and Burnout Related Indicators Using Machine Learning

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DS785 Capstone Project Report

Abstract

Employee attrition poses an ongoing challenge to businesses with the cost of expensive and time-consuming onboarding processes, as well as the loss of talented employees. This study showcases the predictive ability of machine learning models—Logistic Regression, XGBoost, and Random Forest—in predicting employee attrition as well as K-Means segmentation’s ability to enhance predictive performance. This analysis used a synthetic employee burnout and turnover dataset having data on 850,000 individuals, and containing predictor variables related to salary, stress, satisfaction, communication, job level, and performance. This study found that XGBoost was the most effective model at achieving the highest recall (66.0%) and F1 score (43.9%). Decreasing the classification threshold from 0.50 to 0.40 in the XGBoost model increased recall to 91.0% and F1 score to 45.2%, though this came at the cost of a large increase to false positive predictions. K-Means found four clusters with varying attrition rates, but segmentation did not have an improvement on predictive ability. These findings support that modeling could provide assistance in attrition risk profiling, but a low ROC-AUC ceiling and many false positives show limitations to its ability to guarantee prediction results.

1. Problem Definition

Employee attrition is a persistent challenge for businesses as replacing employees requires significant time and money to complete

recruitment, onboarding, training, and time to reach the preferred level of productivity. Being able to predict and flag employees who have a higher predicted likelihood of leaving the company can allow the business to implement targeted retention tactics to both improve employee satisfaction and reduce burnout. In this project I aim to build a model that is best able to balance capturing employees who leave the company with reducing the rate of false positives.

In this project we will be solving a binary classification problem on the target variable `left_company`, which indicates if an employee stayed or left the company. To accomplish this, I will use the classification models Logistic Regression, Random Forest, and XGBoost. I will also perform segmentation analysis using K-Means to give both additional insights into the supervised learning models and perform a descriptive analysis of the segments.

The goal of the project is to acquire a model with the highest achievable recall, F1 score, and ROC-AUC to best support attrition risk screening.

2. Data Collection, Preparation and Cleaning

The dataset used in this project is the Employee Burnout and Turnover Prediction dataset found on Hugging Face. This synthetic dataset contains over 850,000 employee records containing 31 variables that give information on benefits, satisfaction, communication, skills, and job type. The target variable `left_company` was a binary

variable that showed if the employee had left the company.

As this is a synthetic dataset, initial data profiling was done to decide if any of the variables displayed potential data leakage as being either directly tied to the target variable or being unreasonable for an organization to have. Variables that were detected as leakage include: employee_id, burnout_risk, turnover_reason, risk_factor_summary, turnover_probability_generated, persona_name, communication_patterns and role which took our total variable down from 31 to 23.

Variable cleaning and preprocessing was done on the remaining predictor variables. These remaining variables were found to not have any missing values to fix. The remaining fix included grouping employee departments to reduce categorical size, which took us from over 40 departments down to only 17.

3. Exploratory Data Analysis and Feature Engineering

The exploratory data analysis portion of the project focused on understanding variable distributions, with a core focus on relationships between predictor variables and the target variable. The first observation was that the target variable left_company was slightly imbalanced with 71.5% of employees staying with the company and 28.5% of employees leaving. This imbalance of the target variable will need to be addressed during the modeling part of the project.

For the numerical variables in the project most of them were already set to a value between zero and one. The variables that did not fall between zero and one were salary, tenure_months, and overtime hours. Outlier detection was performed on those three variables and while the IQR tests showed that

there were some slight outliers, no variables were removed as the values were still plausible in a real-world setting and removing them would be to remove valuable information. These numerical variables were examined using summary statistics, distribution plots, correlation analysis and comparisons by attrition. The results suggested that attrition was different across many different variables, with some of the most profound differences coming from the variables satisfaction_score and stress_level.

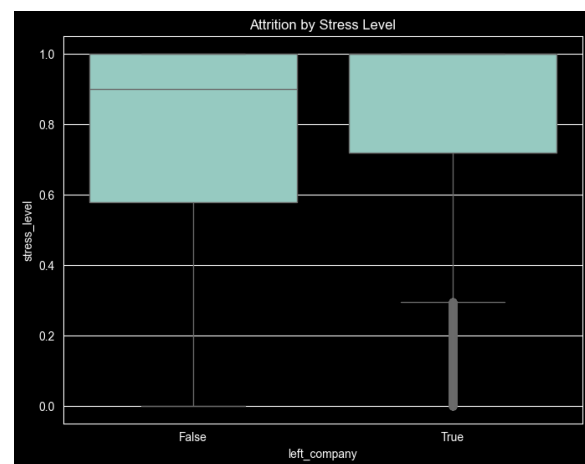


Figure 1: Attrition by stress level

The variables slack_activity, meeting_participation and collaboration_score were found to be duplicates of each other, as well as performance_score and goal_achievement_rate were duplicates of each other. Goal_achievement_rate and collaboration_score were retained while the others were removed. The final correlation analysis showed that none of the remaining predictors were overly correlated with the target variable, with the highest correlation being 0.11.

The categorical variables job_level and department were checked for variation in attrition, the results found that there was nearly

no change in either variable that would significantly separate one department or job from another when based on attrition alone. The variables `technical_skills` and `soft_skills` were found to be highly repeated across all observations and were removed for lack of variation between employees.

Feature creation was used to add additional predictor variables to the dataset. `Feedback_sentiment` was created by performing VADER sentiment analysis on the text variable `recent_feedback` to provide a numerical sentiment score, `recent_feedback` was then removed. `Underpaid_overtime` was created from having salary less than the mean of salary and having overtime hours. `Burnout_indicator_score` was created using `stress_level + workload_score - satisfaction_score`. `New_employee` was created from looking at employees with less than 12 months tenure.

Predictor variable preprocessing was completed using `ColumnTransformer` from `sklearn`. Numeric variables were scaled with `StandardScaler`. This is especially helpful for models like K-Means and Logistic Regression. `Job_level` was given ordinal encoding to maintain the order of Entry, Mid, Senior, Lead and Manager for modeling, and `department` was one hot encoded.

4. Model Development and Evaluation

The modeling section compared results of 3 supervised models: Logistic Regression, Random Forest, and XGBoost. Logistic regression was chosen as an simple baseline model to capture mostly linear relationships. Random Forest and

XGBoost models were selected because they can capture nonlinear relationships in the data. Each of these models are common practice in predicting employee attrition (Iparraguirre-Villanueva et al., 2024; Govindarajan et al., 2025). K-Means clustering was also used as an unsupervised method to determine if different employee segments could be identified, and if those segments could have any impact on improving the results of the supervised learning methods. The dataset was split into an 80-20 test-train set to be used to judge model performance, the dataset was stratified to maintain the same proportion of target variable distribution.

The first baseline Logistic Regression model was trained using no adjustments for class imbalance. This model resulted in an accuracy of 71.5% but 0% for recall, precision, and F1 score, which showed that it simply predicted the most common class for each observation. I then trained a second Logistic Regression model using balanced class weights, this resulted in a decreased accuracy to 52.2%, but increased precision to 32.9%, recall to 64.8%, F1 score to 43.6%, and a ROC-AUC of 58.0%.

Random Forest and XGBoost modeling was also done with methods to handle class imbalance. Random Forest used balanced class weights and XGBoost used `scale_pos_weight` of 2.50 to handle the imbalance. Both models used `RandomizedSearchCV` to perform cross validation and hyperparameter tuning. I performed 5-fold cross validation, with 10 random parameter combinations tested for each, which gave 50 models for each of XGBoost and Random Forest. The cross validation was done to optimize f1 score as it is a balance of recall and precision which are more impactful in this analysis than basic accuracy.

The tuned Random Forest found the best model to use 50 estimators, a maximum depth of 10, minimum sample leaves of 10, minimum sample split of 10, and Gini criterion. The best XGBoost model used 300 estimators, a maximum depth of 3, a learning rate of 0.01, subsample of 1.0, column sample of 0.8, and a minimum child weight of 2. The results showed that XGBoost achieved a better recall and F1 score than the Random Forest model. As these criteria were most important in this project, I selected this to be used for future modeling experiments.

SMOTE oversampling was also tested with an XGBoost model to see if a different way of handling the imbalanced target variable could produce better results. The final XGBoost modeling technique was to perform threshold optimization testing. Thresholds were tested from 0.10 to 0.90, at the default threshold of 0.50 a F1 score of 43.9% was achieved, however the optimal threshold for F1 score was found to be at 0.40 which achieved a score of 45.2%.

Table 1: Model Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	71.5%	0%	0%	0%	58.0%
Balanced Logistic Regression	52.2%	32.9%	64.8%	43.6%	58.0%
Tuned Random Forest	53.0%	33.0%	63.1%	43.4%	58.0%
Tuned XGBoost	51.8%	32.8%	66.0%	43.9%	58.1%
XGBoost + SMOTE	55.4%	32.9%	53.9%	40.8%	56.6%
XGBoost Threshold 0.40	37.0%	30.1%	91.0%	45.2%	58.1%

The results above show the best metrics found for each of the models. XGBoost performed better than both balanced Logistic Regression and Random Forest on recall and F1 scores. XGBoost with SMOTE proved to perform worse than with using scale_pos_weight to handle the class imbalance. Changing the threshold to predict a positive occurrence of the target variable from 0.50 to 0.40 gave us an improved F1 score and a significantly improved recall going up to 91.0% from 66.0%. This came at the cost of a large decrease to accuracy, and a slight decrease to precision. This indicates that while we can flag many of the true leavers of the company, we do end up with significantly more false positives as a result of lowering the threshold. This can be seen in the following confusion matrixes.

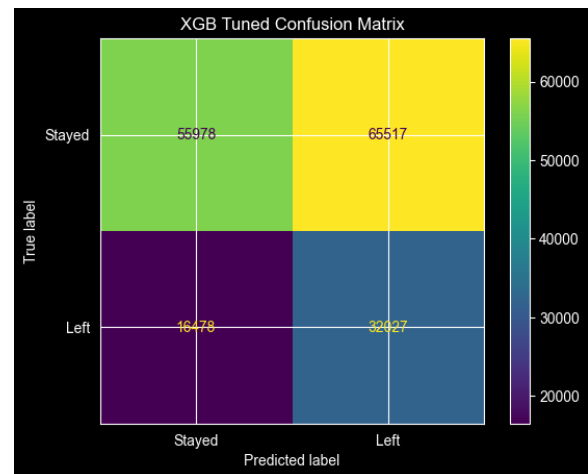


Figure 2: XGBoost Tuned Confusion Matrix

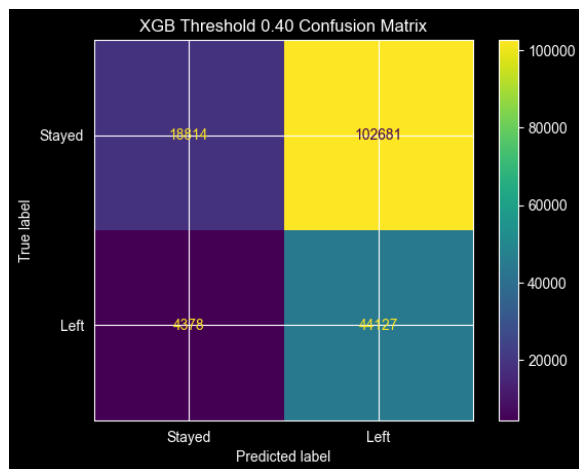


Figure 3: XGBoost Threshold 0.40 Confusion Matrix

Feature importance was pulled out of both the Random Forest and XGBoost models. Each model found that stress_level was the most important predictor variable, followed by burnout_indicator_score. Satisfaction_score and email_sentiment followed up for each model as the next two most important predictors.

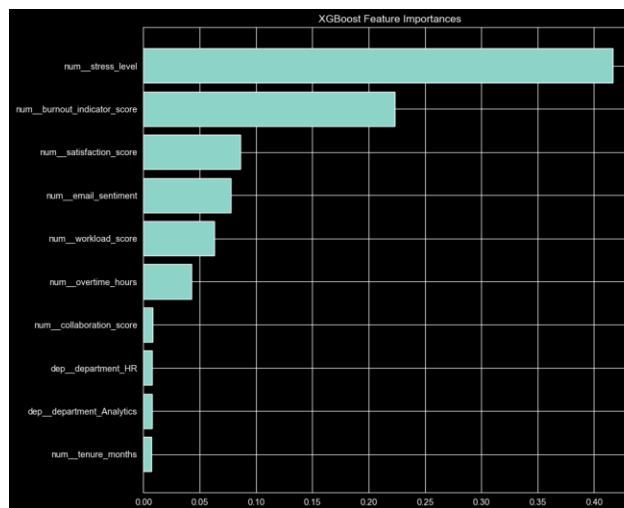


Figure 4: XGBoost Feature Importance Chart

K-Means clustering was performed to view if employee segmentation could provide meaningful outcomes. The elbow method and silhouette score were used to determine four clusters as optimal. The four clusters had

attrition rates ranging from 21.6% to 33.1%. The most significant outcome was that cluster 1 showed the lowest attrition rate and had less than half the stress_level of the other segments, showing that a higher stress level may lead to higher rates of attrition. The cluster labels were applied to the XGBoost modeling and reran; however the results showed no improvement to the model showing that the XGBoost model could already determine many of the reasons that led to the cluster labels outcomes. The clusters were also split into their own dataset and had different XGBoost models tuned on each of them, however the results performed worse than the global XGBoost model. The segmentation results showcased that segments are more beneficial for descriptive analysis rather than predicting outcomes.

5. Results, Insights and Recommendations

The modeling results showed that employee attrition was difficult to accurately predict using the remaining predictor variables. XGBoost performed the best out of the tested models with a recall of 66.0% and an F1 score of 43.9% at the default classification threshold of 0.50. Threshold optimization found an improvement in both recall and F1 score with the best threshold being 0.40. This came with a recall of 91.0% and an F1 score of 45.2%, the downside was that false positives increased significantly with this threshold.

The ROC-AUC score was right around 58.0% for all the models, which shows that the models struggle to separate employees who stayed from those who left. This suggests that employee attrition may be influenced by other

variables not included in the final modeling dataset. This information could have been lost in the removal of leakage variables, as in this synthetic dataset those variables may have more direct relationships with the target variable. Although removing these variables may have reduced the predictive ability of the model, they needed to be removed to ensure realistic model performance.

The K-Means analysis found four employee segments with attrition rates from 21.6% to 33.1%. This shows that the groups identified may have different characteristics leading to different attrition rates. One important finding was that one segment with the lowest attrition rate also had the lowest stress level by over 50% than the other groups. Adding the cluster groups into the supervised modeling process yielded no better results than without them, this indicates that the cluster groups may be better used for descriptive analysis rather than predictive analysis.

Based on the findings here the best model found could be used as a tool for organizations to identify employees at a higher risk of leaving the company. A positive classification is not a guarantee of attrition though as the model does suffer from a high number of false positives. Knowing this, a good use case would be to use the model to flag certain employees as potential leavers, then sending those employees additional information gathering documents such as a survey asking about factors such as satisfaction, stress, and burnout.

6. Conclusion

This project compared supervised and unsupervised machine learning models for

employee attrition on a synthetic dataset. Logistic Regression, Random Forest, and XGBoost were used as supervised models and K-Means was our unsupervised model. XGBoost was found to be the best supervised model and K-Means was determined to be best for descriptive analysis of the determined segments. Overall, the findings suggest that machine learning may be able to support an employee attrition risk screening, but the marginal ROC-AUC results as well as the high false positive rate indicate it should only be used to support decision making, not be used as the final say in predicting employee attrition.

The next steps to be taken in this project would be to use this workflow to test the models on a real-world data set to see if there is differences in both predictive ability as well as variable importance between synthetic and real-world data. In addition, an improvement would be to collect more predictor variables such as exterior market conditions, manager feedback, and quality of benefit packages.

7. References

1. Hugging Face dataset: <https://huggingface.co/datasets/BrotherTony/employee-burnout-turnover-prediction-800k>
2. Govindarajan, R., Komal, K., Sudhakar, R., Pravallika, S., B, D., & G, P. K. (2025). Predicting employee attrition: A comparative analysis of machine learning models using the IBM Human Resource Analytics Dataset. *Procedia Computer Science*, 258, 4084–4093. <https://doi.org/10.1016/j.procs.2025.04.659>
3. Iparraguirre-Villanueva, O., Chauca-Huete, L., Prieto-Chavez, R., & Paulino-Moreno, C. (2024). Employee attrition prediction using machine learning models. *Proceedings of the 22nd LACCEI International Multi-Conference for Engineering, Education and Technology (LACCEI 2024): "Sustainable Engineering for a Diverse, Equitable, and Inclusive Future at the Service of Education, Research, and Industry for a Society 5.0."* <https://doi.org/10.18687/laccei2024.1.1.498>